

Assignment 1 (10% of total marks)

Due date: Sunday, 26 April 2026 by 9:00 pm Singapore time.

Scope:

The tasks of this assignment cover **functional dependency**. The assignment covers the topics discussed in lecture 1, and 2.

Assessment criteria:

Marks will be awarded for:

- Correct,
- Comprehensive, and
- Appropriate

application of the materials covered in this subject.

A late submission penalty (5% of the total mark) will be applied for every 24 hours late.

Submission must be done online by using the submission link associated with assignment 1 for this subject on MOODLE. The size limit for all submitted materials is 20MB.

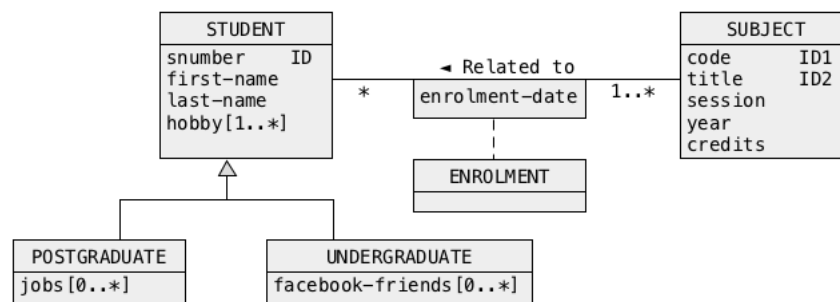
Note: Only one submission is allowed. Ensure you upload the correct files with the exact required contents before the deadline. If you need an extension (maximum 1 week), you must submit an Academic Consideration request via SOLS **before or on the due date**. Requests submitted after the due date are likely to be rejected.

Assignment Specification:

Task 1 (3.0 marks)

Analysis of relational schemas and normalization

Consider the following conceptual schema of a sample database domain that contains information about undergraduate and postgraduate students, subjects, and enrolments performed by the students.



A database designer applied a process of logical design and he/she transformed the conceptual schema into the following collection of relational schemas.

STUDENT(snumber, first-name, last-name, hobby, jobs, facebook-friends)
primary key = (snumber)

SUBJECT(code, title, session, year, credits)
primary key = (code)
candidate key = (title)

ENROLMENT(snumber, code, title, enrolment-date)
primary key = (snumber, code, enrolment-date)
foreign key 1 = (snumber) references STUDENT(snumber)
foreign key 2 = (code) references SUBJECT(code)
foreign key 3 = (title) references SUBJECT(title)

- (1) Find all functional and multivalued dependencies in the relational schemas STUDENT, SUBJECT, and ENROLMENT.
- (2) Find all minimal keys in the relational schemas STUDENT, SUBJECT, and ENROLMENT. List the derivations of all minimal keys.
- (3) For each one of the relational schemas find the highest normal form a schema is in. List the justifications for each highest normal form found.
- (4) Decompose all relational schemas that are not in 4NF into 4NF. List all relational schemas obtained from the decompositions.

Deliverables

A file solution1.pdf with the outcomes of the steps (1), (2), (3), and (4) listed above. Note, that "educated guesses" of the solutions score no marks. You must provide the complete justifications of your answers.

Submission of a file with a different name and/or different extension and/or different type scores no marks

Task 2 (3.0 marks)

Analysis of relational schemas and normalization

Consider the relational schemas given below and the respective sets of functional dependencies valid in the schemas.

$R(P, Q, R, S, T, U, V, W)$

Functional Dependency: $RW \rightarrow V, P \rightarrow QR, Q \rightarrow RUW, T \rightarrow P, U \rightarrow TV$

- (i) Find **all** the minimal super keys of the relational table R . List the derivations of all minimal keys.
- (ii) Identify the highest normal form of the relational table R . Remember that the identification of a normal form requires analysis of the valid functional dependencies.
- (iii) Decompose the relational table R into minimal number of normalized relational tables in BCNF. Remember to indicate the primary key and foreign keys (if any).

Deliverables

A file **solution1.pdf** with the outcomes of the steps (i), (ii), and (iii) listed above. Note, that "educated guesses" of the solutions score no marks. You must provide the complete justifications of your answers.

Submission of a file with a different name and/or different extension and/or different type scores no marks!

Task 3 (3.0 marks)

Analysis of relational schemas and normalization

A book is described by a catalog number, a book title, author's first name, author's last name, a publisher name, the year the book was published, and a price of the book. An author may write many books and each book may be written by one or more authors. The information is stored in a relational table book as follow:

BOOK (CatalogNum, BookTitle, AuthorFName, AuthorLName, PublisherName, YearOfPublication, Price)

The following dependencies exist in the relational table BOOK:

- *CatalogNum* → *BookTitle, PublisherName, YearOfPublication*
- *BookTitle, PublisherName, YearOfPublication* → *Price*
- *AuthorFirstName, AuthorLastName, BookTitle* → *CatalogNum*

- Find all the minimal super key of the relational table BOOK. List the derivations of all minimal keys.
- Identify the highest normal form of the relational table BOOK. List the justifications for each highest normal form found.
- Decompose the relational table BOOK into minimal number of relational tables in BCNF. List all relational tables obtained from the decompositions.

Deliverables

A file solution1.pdf with the outcomes of the steps (i), (ii), and (iii) listed above. Note, that "educated guesses" of the solutions score no marks. You must provide the complete justifications of your answers.

Submission of a file with a different name and/or different extension and/or different type scores no marks!

Submissions

This assignment is due on Sunday, 26 April 2026 by 9:00 pm Singapore time.

Submit the files **solution1.pdf**, **solution2.pdf**, and **solutions3.pdf** through Moodle in the following way:

- 1) Zip all the files (Solution1.pdf, solution2.pdf, and solution3.pdf into one zipped folder.)
- 2) Access Moodle at **<http://moodle.uowplatform.edu.au/>**
- 3) To log in use a Login link located in the right upper corner the Web page or in the middle of the Web page
- 4) When successfully logged in, select a site CSCI235 (SP226) Database Systems
- 5) Scroll down to a section Submissions of Assignments
- 6) Click at Submit Your Assignment 1 here link.
- 7) Click at the button Add Submission
- 8) Move the zipped file created in Step 1 above into an area provided in Moodle. You can drag and drop files here to add them. You can also use a link *Add...*
- 9) Click at a button Save changes,
- 10) Click at check box to confirm authorship of a submission,
- 11) When you are satisfied, remember to click at a button Submit assignment.

A policy regarding late submissions is included in the subject

outline. Only one submission per student is accepted.

Assignment 1 is an individual assignment, and it is expected that all its tasks will be solved individually without any cooperation with the other students. Plagiarism is treated seriously. Students involved will likely receive zero. If you have any doubts, questions, etc. please consult your lecturer or tutor during lab classes or over e-mail.

End of specification